

and so on, with some of the numbers rounded off for convenience. You'll notice that the slash indicates division. For example, $f/16$ means that the aperture is equal to the focal length divided by sixteen; that is, if the camera has an 80-mm lens, the light that reaches the film arrives through an opening that is 5 mm ($80 \text{ mm}/16$) in diameter. Given below is a table of aperture stops of a typical 50-mm lens.

f/stop	Diameter of aperture (mm)	Radius of aperture (mm)	Area of Aperture (sq. mm)
f/1.0	50.0	25.0	1,963
f/1.4	35.7	17.9	1,002
f/2.0	25.0	12.5	491
f/2.8	17.9	8.9	250
f/4	12.5	6.3	123
f/5.6	8.9	4.5	63
f/8	6.3	3.1	31
f/11	4.5	2.3	16
f/16	3.1	1.6	8
f/22	2.3	1.1	4
(As shown on screen/ lens)	(50 mm divided by f/stop)	(1/2 the diameter)	(pi X the radius squared)

As the f-number increases, the diameter of the aperture decreases, letting in half the amount of light as the previous one. Effectively, if you want to *open up* in photography jargon, then you'll be reducing the f-number. Similarly, *stopping down* in shutterbug lingo would mean you're constricting the opening by increasing the f-number (and thereby acquiring a sharper overall image).

Lenses are usually described by their maximum possible aperture (represented by the smallest f-stop number). When in Aperture Priority mode, a high f-number refers to a slower shutter speed. A low f-number like $f/1.4$ will automatically speed up the shutter speed when in Aperture Priority Mode to avoid over-exposure.